

Amplitude and Angle Modulation Formulae

1 Basic Signal Definitions

- Message Signal: $m(t)$
- Carrier Signal: $c(t) = A \cos(2\pi f_c t)$

Modulation is the process of varying a characteristic of the carrier (amplitude, frequency, or phase) according to $m(t)$. For Amplitude Modulation, the carrier amplitude A varies proportionally to $m(t)$.

2 Amplitude Modulation (AM) Formulae

2.1 Basic AM (Conventional AM)

- Modulated Signal:

$$x(t) = (A + m(t)) \cos(2\pi f_c t)$$

- Frequency Spectrum:

$$X(f) = \frac{A}{2} [\delta(f + f_c) + \delta(f - f_c)] + \frac{1}{2} [M(f + f_c) + M(f - f_c)]$$

- Carrier Power:

$$P_c = \frac{A^2}{2}$$

- Sideband Power:

$$P_s = \frac{1}{2} P_m$$

- Total Power:

$$P_{AM} = P_c + P_s = \frac{A^2}{2} + \frac{P_m}{2}$$

- Power Efficiency:

$$\eta = \frac{P_s}{P_s + P_c} \times 100\% = \frac{P_m}{A^2 + P_m} \times 100\%$$

- Modulation Index (μ):

$$\mu = \frac{m_p}{A}, \quad \mu = \frac{x_{\max} - x_{\min}}{x_{\max} + x_{\min}} \quad (\text{from envelope})$$

- Bandwidth:

$$BW = 2B$$

2.2 Double Sideband-Suppressed Carrier (DSB-SC)

- Modulated Signal:

$$x(t) = m(t) \cos(2\pi f_c t)$$

- Auto-Correlation Function:

$$R_x(\tau) = \frac{1}{2} R_m(\tau) \cos(2\pi f_c \tau)$$

- Power Spectral Density:

$$S_x(f) = \frac{1}{4} [S_m(f - f_c) + S_m(f + f_c)]$$

- Average Power:

$$P_{DSB-SC} = \frac{1}{2} P_m$$

- Bandwidth:

$$BW = 2B$$

2.3 Single Sideband (SSB)

- Hilbert Transform:

$$X_h(f) = -j \operatorname{sgn}(f) X(f)$$

- Upper SSB (USSB) Signal:

$$x_{USSB}(t) = m(t) \cos(2\pi f_c t) - m_h(t) \sin(2\pi f_c t)$$

- Lower SSB (LSSB) Signal:

$$x_{LSSB}(t) = m(t) \cos(2\pi f_c t) + m_h(t) \sin(2\pi f_c t)$$

- Average Power:

$$P_{SSB} = \frac{P_m}{2} + \frac{P_{m_h}}{2} = P_m$$

- Bandwidth:

$$BW = B$$

3 Angle Modulation

3.1 General Angle Modulation

$$x(t) = A \cos(2\pi f_c t + \phi(t))$$

- Instantaneous Frequency:

$$f_i(t) = f_c + \frac{1}{2\pi} \frac{d\phi(t)}{dt}$$

- Instantaneous Frequency Deviation:

$$\Delta f(t) = f_i(t) - f_c = \frac{1}{2\pi} \frac{d\phi(t)}{dt}$$

- Average Transmitted Power:

$$P = \frac{A^2}{2}$$

3.2 Frequency Modulation (FM)

- Instantaneous Frequency:

$$f_i(t) = f_c + k_f m(t)$$

- Phase:

$$\phi(t) = 2\pi k_f \int_{-\infty}^t m(\tau) d\tau$$

- FM Signal:

$$x_{FM}(t) = A \cos \left(2\pi f_c t + 2\pi k_f \int_{-\infty}^t m(\tau) d\tau \right)$$

- Maximum Frequency Deviation:

$$\Delta f = k_f m_p$$

- Narrowband FM Approximation:

$$x_{NBFM}(t) \approx A \left[\cos \omega_c t - k_f \left(\int_{-\infty}^t m(\tau) d\tau \right) \sin \omega_c t \right]$$

- Modulation Index:

$$\beta = \frac{\Delta f}{f_m}$$

- Carson's Rule:

$$B_{FM} = 2(\Delta f + B) = 2(\beta + 1)f_m$$

3.3 Phase Modulation (PM)

- Phase:

$$\phi(t) = k_p m(t)$$

- PM Signal:

$$x_{PM}(t) = A \cos(2\pi f_c t + k_p m(t))$$

- Maximum Phase Deviation:

$$\Delta \phi = k_p m_p$$

- Narrowband PM Approximation:

$$x_{NBPM}(t) \approx A \left[\cos \omega_c t - k_p m(t) \sin \omega_c t \right]$$